

Daniel Vanegas Forero

Geomatics Engineer | GIS & Data Engineer

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🐙 [GitHub/DanielVanegas](https://github.com/DanielVanegas) | 🌐 Portuguese citizen | 📍 Based in Porto, Portugal

Geomatics Engineer with hands-on experience designing and maintaining geospatial data pipelines and automated workflows using Python, SQL, and PostGIS. I combine GIS production (QGIS/ArcGIS), ETL processes (FME), and spatial analytics and econometric modeling applied to land administration, property valuation, and territorial analysis. I deliver reproducible, efficient workflows, decision-ready maps, and measurable improvements in data processing performance. Interested in data-driven geospatial and environmental applications. Portuguese citizen, legally authorized to work in the European Union.

Work experience

Feb 2025 - Present **ESC Ingenieros** **Property Valuer**

- Development of internal systems for managing property appraisals and technical visits, automating the organization of case files and report generation.
- Design and implementation of geospatial ETL pipelines using Python and PostgreSQL/PostGIS, reducing data processing time by over 50% and improving traceability and reproducibility.
- Automated generation of contextual and analytical maps from field-collected coordinates for report-ready outputs.
- Portfolio: [procesador_visitas](#), [form_visitas](#)

Sep 2024 - Oct 2025 **Tinsa Colombia** **Property Valuer**

- Preparation of property valuation reports in compliance with technical standards.
- Coordination of technical field visits and adherence to delivery deadlines.
- Featured project: development of a Python desktop application (PyQt6 + xlwings) to automate the transfer and validation of data and images in Excel, reducing manual errors and improving traceability.
- Portfolio: [pyqt-excel-automation](#)

Sep 2024 - Dec 2024 **Cadastral Data Technician** **Data Tools S.A.S.**

- Provision of professional services as a Cadastral Data Technician in the cadastral update process for the municipality of Montería - Colombia. Responsibilities included the digitalization, verification, and organization of field-collected data, ensuring data quality and consistency with original documents in compliance with technical and regulatory standards.
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Jul 2024 - Oct 2024
Cadastral Analyst
Realtix S.A.S.

- Participation in the multipurpose cadastral update process, including data collection, technical analysis, parcel digitization, layer and database updating, quality control, and production of cartographic outputs.
 - Developed Python (arcpy) modules to automatically validate and correct attributes according to the LADM-COL model, ensuring consistency among parcels, buildings, and property units.
 - Implemented quality-control routines that reduced database errors and optimized parcel validation time.
 - Portfolio: [validador_ladm](#)
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Education

B.Sc. in Cadastral and Geodesy Engineering

Universidad Distrital Francisco José de Caldas (Colombia)

February 2017 - June 2024

Thesis: Spatial econometric modeling of apartment unit values in Soacha (2017–2023).

Applied ESDA (global Moran, LISA), Lagrange multiplier tests, and Spatial Error and Lag Models (SEM/SLM) for valuation analysis.

Automated dataset creation through PDF extraction using Power Automate + AI Builder and validation in Excel/QGIS/PostGIS.

Technical Skills

- Programming & Data: Python, SQL
 - Geospatial Databases: PostgreSQL, PostGIS
 - GIS & Spatial Analysis: QGIS, ArcGIS, spatial analysis, cartography
 - ETL & Automation: FME, geopandas, pandas, arcpy, GDAL
 - Dev & Environment: Git, GitHub/GitLab, Linux
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Languages

- Spanish (native)
 - English (B1)
 - Portuguese (A2)
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Featured Projects

Real Estate Scraper – PostgreSQL & Python

- Developed an automated system to extract and store real estate listings from FincaRaiz into a PostgreSQL database.
 - Implemented validation processes, status updates, and link failure detection with concurrent execution in Python.
 - Code available at: [scraping_fr](#)
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